Experiment No. 3
Explore Inferential Statistic on the given dataset
Date of Performance: 10/02/2025
Date of Submission: 24/02/2025

Academic Year: 2024-25 **Semester: VIII**

Class / Branch: BE Computer **Subject: Applied Data Science Lab**

Experiment No. 3

Aim: To explore the inferential statistic t-test on the given dataset.

Dataset: In this experiment, three fictitious datasets are used. Reliance data mart dataset is used to explore One sample t-test. Crocin dataset and Prescore-Postscore datasets are used to study Two sample paired t-test.

1. **Software used:** Google Colaboratory / Jupyter Notebook

2. **Theory :-**

- Z-Test & T-Tests are Parametric Tests, where the Null Hypothesis is less than, greater than or equal to some value.
- A z-test is used if the population variance is known, or if the sample size is larger than 30, for an unknown population variance.
- If the sample size is less than 30 and the population variance is unknown, we must use a t-test.

T test is a type of inferential statistic used to study if there is a statistical difference between two groups. Mathematically, it establishes the problem by assuming that the means of the two distributions are equal (H_0: $\mu_1=\mu_2$). If the t-test rejects the null hypothesis (H_0: $\mu_1=\mu_2$), it indicates that the groups are highly probably different.

The statistical test can be one-tailed or two-tailed. The **one-tailed test** is appropriate when there is a difference between groups in a specific direction. It is less common than the **two-tailed test.** When choosing a *t* test, you will need to consider two things: whether the groups being compared come from a single population or two different populations, and whether you want to test the difference in a specific direction.

There are three main types of t-test :

- **One Sample t-test :** Compares **mean of a single group against a known/**hypothesized/ population mean.
- **Two Sample: Paired Sample T Test:** Compares means from the **same group at different times.**
- **Two Sample: Independent Sample T Test:** Compares means for **two different groups.**

One Sample t-test:

$$t = \frac{\text{(Sample Mean} - \text{Population Mean)}}{\text{Standard Error}}$$

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

$\bar{x}$ Sample mean
μ Population mean
s Sample standard deviation
n Sample size

Two-sample - Paired Sample t-test

$$t = \frac{\bar{d}}{s / \sqrt{n}}$$

$\bar{d}$ =Mean of the difference
s=Standard deviation of the difference
n =is the sample size (i.e., size of d)

- If the **calculated t value is less than critical t value or greater that the critical value** (obtained from a critical value table called the T-distribution table) **then reject the null hypothesis.**
- **P-value <significance level (α) => Reject your null hypothesis** in favor of your alternative hypothesis. Your result is statistically significant.
- **P-value >= significance level (α) => Fail to reject your null hypothesis.** Your result is not statistically significant

Program:
https://colab.research.google.com/drive/1N1v4NUxQ1BJNgh_OXpgF20UN_vb5PaLn

Output:

```
Initial Data:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29 entries, 0 to 28
Data columns (total 1 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Rice_Bag_Weight  29 non-null     float64
dtypes: float64(1)
memory usage: 364.0 bytes
None
    Rice_Bag_Weight
0              24.5
1              24.7
2              25.6
3              25.0
4              24.7
5              23.3
6              23.3
7              24.0
8              25.1
9              24.3
10             23.3
```

```
One-Sample t-test Results:
T-Statistic: -5.236976851657697
P-Value: 1.4512165732179395e-05
Result: Reject the null hypothesis (Significant difference found).
```

```
Descriptive Statistics:
Mean: 24.446206896551722
Median: 24.5
Mode: 24.7
Max Value: 25.6
Min Value: 23.3
Standard Deviation: 0.569463493307988
Standard Error: 0.10574671592695357
Kurtosis: -0.1784721078672531
Skewness: -0.38055508159392537
Range: 2.3000000000000007
Sum: 708.9399999999999
Count: 29

One-Sample t-test Results:
T-Statistic: -5.236976851657697
P-Value: 1.4512165732179395e-05
T Critical Value: 2.048407141795244
Result: Reject the null hypothesis (Significant difference found).
```

```
Pre-Post Score Data:
   Pre_Score Post_Score  Diff  Unnamed: 3  Unnamed: 4 Unnamed: 5  Unnamed: 6
0       18.0         22  -4.0         NaN         NaN        NaN         NaN
1       21.0         25  -4.0         NaN         NaN        NaN         NaN
2       16.0         17  -1.0         NaN         NaN        NaN         NaN
3       22.0         24  -2.0         NaN         NaN        NaN         NaN
4       19.0         16   3.0         NaN         NaN        NaN         NaN

Paired Sample T-Test Results:
Mean Pre-Score: 18.4
Mean Post-Score: 20.45
Variance Pre-Score: 9.936842105263159
Variance Post-Score: 16.471052631578946
Number of Observations: 20
Pearson Correlation: 0.7174770398523397
Hypothesized Mean Difference: 0
Degrees of Freedom: 19
T-Statistic: -3.231252665580312
P-Value (One-Tail): 0.002197482996592832
T Critical Value (One-Tail): 1.729132811521367
P-Value (Two-Tail): 0.004394965993185664
T Critical Value (Two-Tail): 2.093024054408263
Result: Reject the null hypothesis (Significant difference found).
```

Conclusion :- One sample t-test has been done on the reliance data mart dataset and it has been found that difference exists between the rice bag population mean and rice bag sample mean. Two sample paired t-test has been done on the prescore-post score dataset and Crocin dataset. In the prescore-post score dataset difference exists between the mean pre-score before studying the module and mean prescore after studying the module. In the crocin dataset it is found that temperature difference exists before and after having the crocin tablet.